

Estimation Theory

Estimation Problem

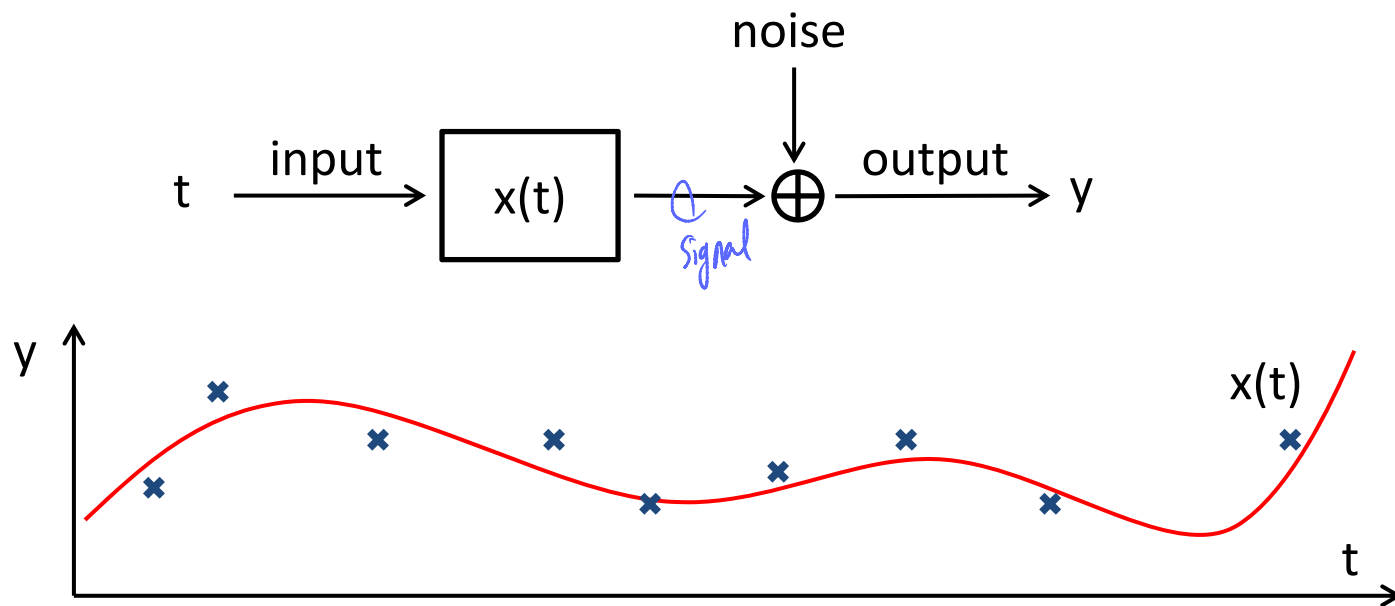
Bayesian Filtering

Kalman Filters – A Special Case of Bayesian Filtering

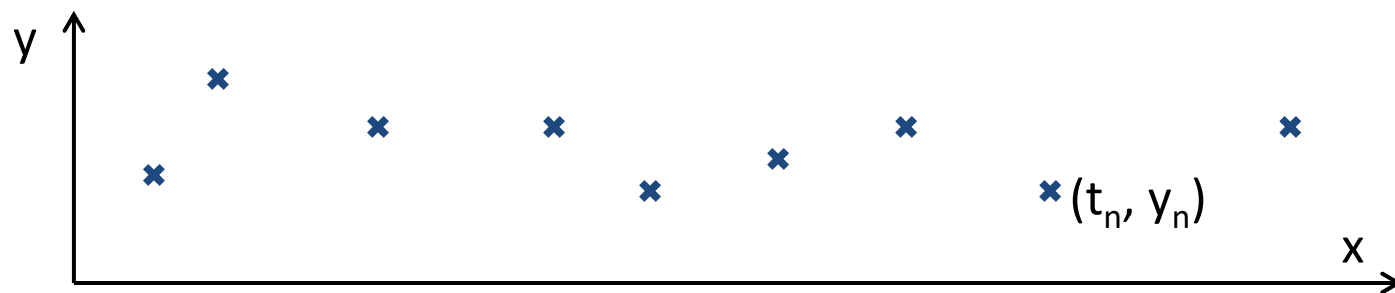
Other Filtering Algorithms: Extended Kalman Filters, Particle Filters

BAYESIAN FILTERING

Estimation Problem



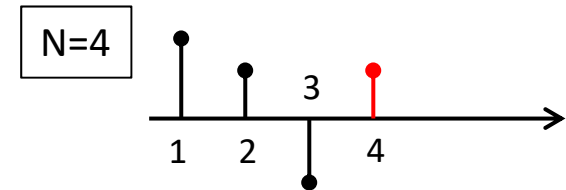
Estimation problem: Find $x(t)$ from data $\{(t_n, y_n) : n=1, \dots, N\}$



Estimation Problems

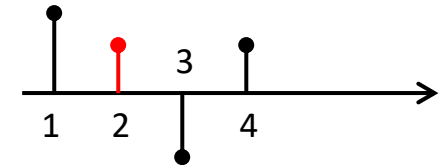
- Filtering

- Estimate $x(t_N)$ from $\{(t_n, y_n) : n=1, \dots, N\}$



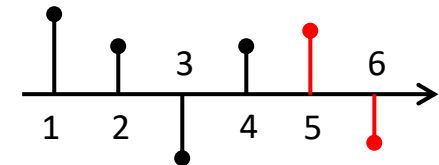
- Smoothing

- Estimate $x(t_i)$ from $\{(t_n, y_n) : n=1, \dots, N\}$, $1 \leq i \leq N$

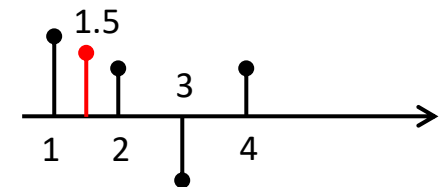


- Prediction

- Estimate $x(t_m)$ from $\{(t_n, y_n) : n=1, \dots, N\}$, $m > N$



- Interpolation



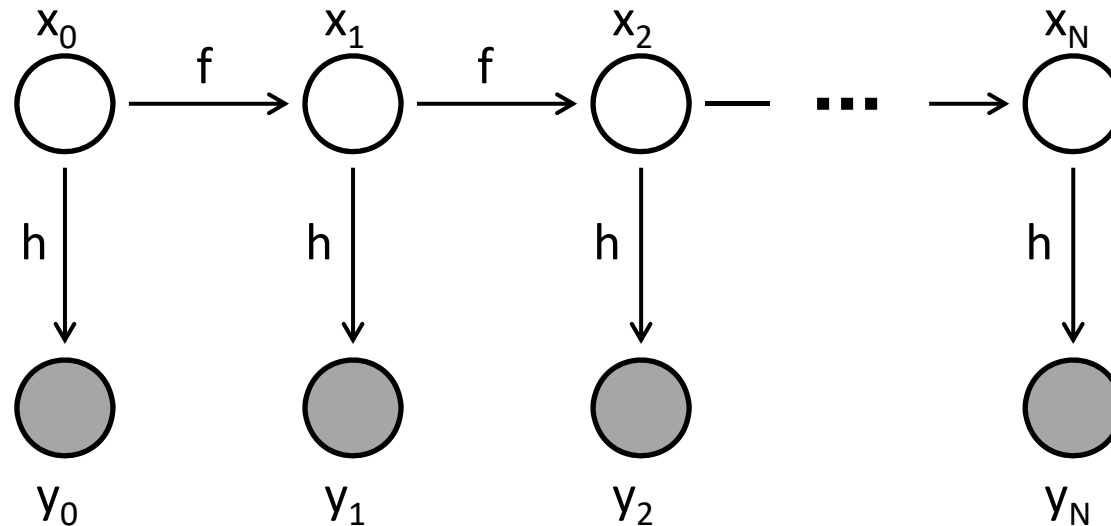
Dynamical Systems

Discrete-time dynamic and measurement model

$$\begin{aligned}x_{t+1} &= f(x_t) + u_t \\ y_t &= h(x_t) + w_t\end{aligned}$$

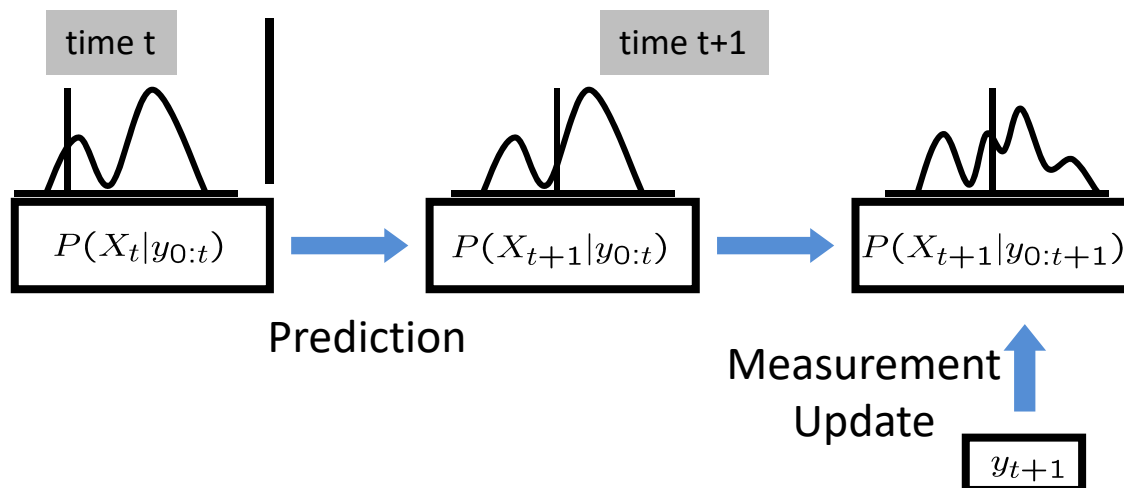
- x_t : state at time t
- y_t : observation at time t
- u_t, w_t : white noise processes

Graphical model:



Bayesian Filtering

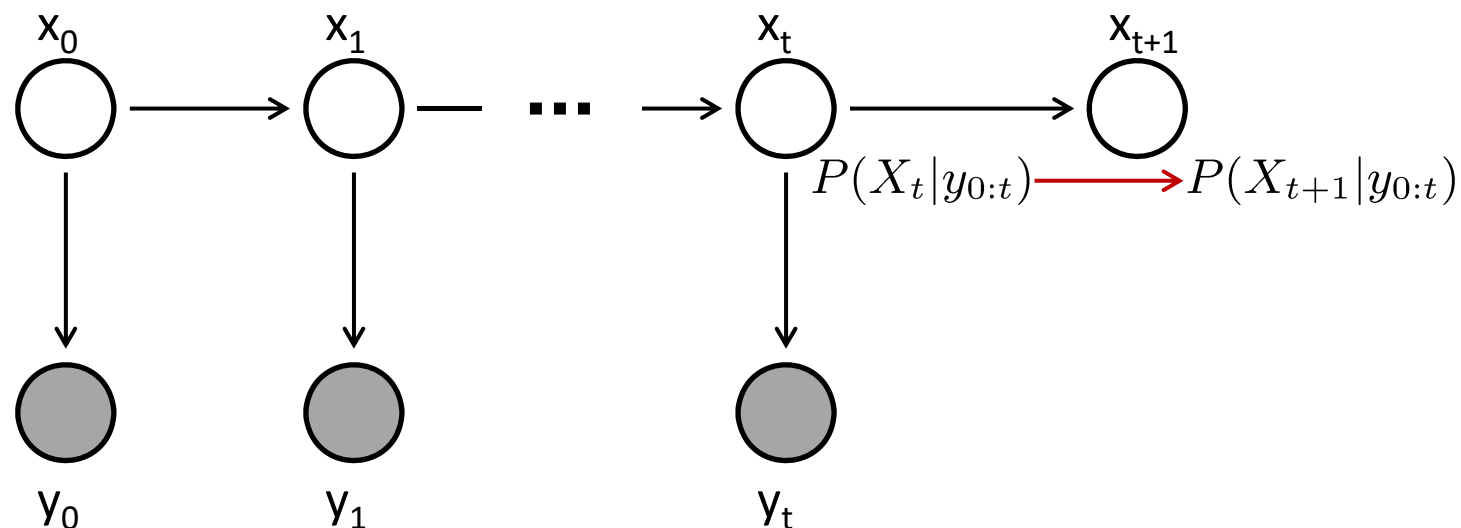
- Compute the posterior $P(X_t|y_{0:t})$ for each t , where $y_{0:t} = \{y_0, y_1, \dots, y_t\}$.
- It is assumed that the prior $P(X_0)$ is known.
- Bayesian filtering is done in two steps: (1) prediction and (2) measurement update.
- Bayesian filtering algorithm
 1. Compute $P(X_{t+1}|y_{0:t})$ from $P(X_t|y_{0:t})$ (prediction)
 2. New observation y_{t+1}
 3. Compute $P(X_{t+1}|y_{0:t+1})$ (measurement update)
 4. Go to the next time



Examples of Bayesian filtering:

- Hidden Markov model, *← State is discrete, finite*
- Kalman filters, *← State is continuous*
- Particle filters, *← State is continuous*
- ...

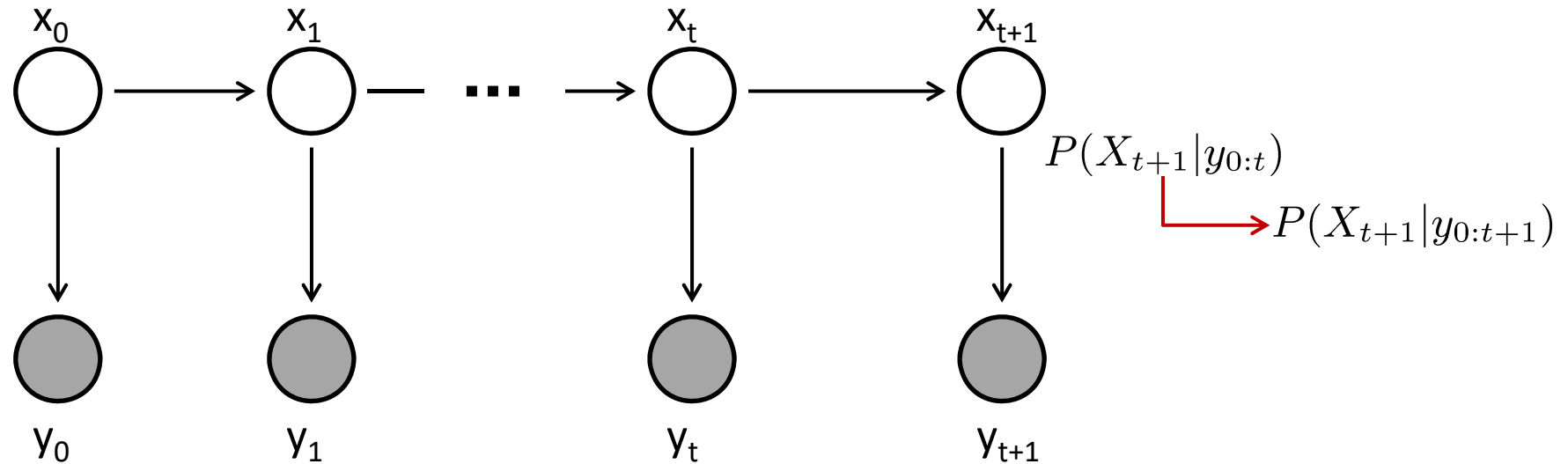
Prediction



- $P(X_t | y_{0:t})$ is available from the previous time step.
- Prediction (or dynamic update):

$$\begin{aligned} P(X_{t+1} | y_{0:t}) &= \int P(X_{t+1} | x_t, y_{0:t}) P(x_t | y_{0:t}) dx_t \\ &= \int \underbrace{P(X_{t+1} | x_t)}_{\text{Dynamic Model.}} P(x_t | y_{0:t}) dx_t \end{aligned}$$

Measurement Update



- New measurement y_{t+1} .
- Measurement update (Bayes rule):

$$\begin{aligned}
 P(X_{t+1} | y_{0:t+1}) &= \frac{P(y_{t+1} | X_{t+1}, y_{0:t}) P(X_{t+1} | y_{0:t})}{\int P(y_{t+1} | x_{t+1}, y_{0:t}) P(x_{t+1} | y_{0:t}) dx_{t+1}} \\
 &= \frac{P(y_{t+1} | X_{t+1}) P(X_{t+1} | y_{0:t})}{\int P(y_{t+1} | x_{t+1}) P(x_{t+1} | y_{0:t}) dx_{t+1}}
 \end{aligned}$$

A Special Case of Bayesian Filtering

KALMAN FILTERING

Linear Gaussian Model

- Dynamic model

$$x_{t+1} = Ax_t + Gw_t$$

- Measurement model

$$y_t = Cx_t + v_t$$

- Initial state distribution: $x_0 \sim \mathcal{N}(\mu_0, \Sigma_0)$ (assume $\mu_0 = 0$ for now)
- Noises: $w_t \sim \mathcal{N}(0, Q)$, $v_t \sim \mathcal{N}(0, R)$. w_t, v_t , and x_0 are independent.
- Unconditional distribution of x_t (i.e., y_t are not observed)

$$\begin{aligned}\mathbb{E}(x_t) &= 0 \quad \forall t \\ \mathbf{var}(x_{t+1}) &= \Sigma_{t+1} = \mathbb{E}(x_{t+1}x_{t+1}^T) \\ &= \mathbb{E}\left((Ax_t + Gw_t)(Ax_t + Gw_t)^T\right) \\ &= A\mathbb{E}(x_t x_t^T)A^T + G\mathbb{E}(w_t w_t^T)G^T \\ &= A\Sigma_t A^T + GQG^T\end{aligned}$$

Filtering

- Dynamic update

$$P(x_t|y_0, \dots, y_t) \rightarrow P(x_{t+1}|y_0, \dots, y_t) = \int P(x_{t+1}|x_t)P(x_t|y_0, \dots, y_t)dx_t$$

- Measurement update *Bayes' Rule*

$$P(x_{t+1}|y_0, \dots, y_t) \rightarrow P(x_{t+1}|y_0, \dots, y_t, y_{t+1}) \propto P(y_{t+1}|x_{t+1})P(x_{t+1}|y_0, \dots, y_t)$$

- $P(x_{t+1}, y_{t+1}|y_0, \dots, y_t) \sim \mathcal{N}(\mu, \Sigma)$, where

$$\mu = \begin{pmatrix} \mu_x \\ \mu_y \end{pmatrix}, \quad \Sigma = \begin{pmatrix} \Sigma_{xx} & \Sigma_{xy} \\ \Sigma_{yx} & \Sigma_{yy} \end{pmatrix}$$

Conditional PDF of Multivariate Gaussian

Theorem 10.2 (Conditional PDF of Multivariate Gaussian) *If $\mathbf{x}$ and $\mathbf{y}$ are jointly Gaussian, where $\mathbf{x}$ is $k \times 1$ and $\mathbf{y}$ is $l \times 1$, with mean vector $[E(\mathbf{x})^T E(\mathbf{y})^T]^T$ and partitioned covariance matrix*

$$\mathbf{C} = \begin{bmatrix} \mathbf{C}_{xx} & \mathbf{C}_{xy} \\ \mathbf{C}_{yx} & \mathbf{C}_{yy} \end{bmatrix} = \begin{bmatrix} k \times k & k \times l \\ l \times k & l \times l \end{bmatrix} \quad (10.23)$$

so that

$$p(\mathbf{x}, \mathbf{y}) = \frac{1}{(2\pi)^{\frac{k+l}{2}} \det^{\frac{1}{2}}(\mathbf{C})} \exp \left[-\frac{1}{2} \left(\begin{bmatrix} \mathbf{x} - E(\mathbf{x}) \\ \mathbf{y} - E(\mathbf{y}) \end{bmatrix} \right)^T \mathbf{C}^{-1} \left(\begin{bmatrix} \mathbf{x} - E(\mathbf{x}) \\ \mathbf{y} - E(\mathbf{y}) \end{bmatrix} \right) \right],$$

then the conditional PDF $p(\mathbf{y}|\mathbf{x})$ is also Gaussian and

$$E(\mathbf{y}|\mathbf{x}) = E(\mathbf{y}) + \mathbf{C}_{yx} \mathbf{C}_{xx}^{-1} (\mathbf{x} - E(\mathbf{x})) \quad (10.24)$$

$$\mathbf{C}_{y|x} = \mathbf{C}_{yy} - \mathbf{C}_{yx} \mathbf{C}_{xx}^{-1} \mathbf{C}_{xy}. \quad (10.25)$$

Filtering

- Dynamic update

$$P(x_t|y_0, \dots, y_t) \rightarrow P(x_{t+1}|y_0, \dots, y_t) = \int P(x_{t+1}|x_t)P(x_t|y_0, \dots, y_t)dx_t$$

- Measurement update

$$P(x_{t+1}|y_0, \dots, y_t) \rightarrow P(x_{t+1}|y_0, \dots, y_t, y_{t+1}) \propto P(y_{t+1}|x_{t+1})P(x_{t+1}|y_0, \dots, y_t)$$

- $P(x_{t+1}, y_{t+1}|y_0, \dots, y_t) \sim \mathcal{N}(\mu, \Sigma)$, where

$$\mu = \begin{pmatrix} \mu_x \\ \mu_y \end{pmatrix}, \quad \Sigma = \begin{pmatrix} \Sigma_{xx} & \Sigma_{xy} \\ \Sigma_{yx} & \Sigma_{yy} \end{pmatrix}$$

- $P(x_{t+1}|y_0, \dots, y_t, y_{t+1}) \sim \mathcal{N}(\mu_{x|y}, \Sigma_{x|y})$, where

$$\begin{aligned} \mu_{x|y} &= \mu_x + \Sigma_{xy}\Sigma_{yy}^{-1}(y - \mu_y) \\ \Sigma_{x|y} &= \Sigma_{xx} - \Sigma_{xy}\Sigma_{yy}^{-1}\Sigma_{yx} \end{aligned}$$

Dynamic Update

$$\begin{array}{ll}
 \hat{x}_{t|t} &:= \mathbb{E}(x_t | y_0, \dots, y_t) \\
 P_{t|t} &= \mathbb{E}(\tilde{x}_{t|t} \tilde{x}_{t|t}^T | y_0, \dots, y_t) \\
 \tilde{x}_{t|t} &= x_t - \hat{x}_{t|t}
 \end{array}
 \quad \Longrightarrow \quad
 \begin{array}{ll}
 \hat{x}_{t+1|t} &:= \mathbb{E}(x_{t+1} | y_0, \dots, y_t) \\
 P_{t+1|t} &= \mathbb{E}(\tilde{x}_{t+1|t} \tilde{x}_{t+1|t}^T | y_0, \dots, y_t) \\
 \tilde{x}_{t+1|t} &= x_{t+1} - \hat{x}_{t+1|t}
 \end{array}$$

$$\begin{aligned}
 \hat{x}_{t+1|t} &= \mathbb{E}(x_{t+1} | y_0, \dots, y_t) \\
 &= \mathbb{E}(Ax_t + Gw_t | y_0, \dots, y_t) \\
 &= A\hat{x}_{t|t}
 \end{aligned}
 \qquad x_{t+1} = Ax_t + Gw_t$$

$$\begin{aligned}
 P_{t+1|t} &= \mathbb{E}[\tilde{x}_{t+1|t} \tilde{x}_{t+1|t}^T | y_0, \dots, y_t] \\
 &= \mathbb{E}[(Ax_t + Gw_t - A\hat{x}_{t|t})(Ax_t + Gw_t - A\hat{x}_{t|t})^T | y_0, \dots, y_t] \\
 &= \mathbb{E}[(A(x_t - \hat{x}_{t|t}) + Gw_t)(A(x_t - \hat{x}_{t|t}) + Gw_t)^T | y_0, \dots, y_t] \\
 &= AP_{t|t}A^T + GQG^T
 \end{aligned}$$

$\mathbb{E}((x_t - \hat{x}_{t|t})(x_t - \hat{x}_{t|t})^T) := P_{t|t}$

Measurement Update (1)

$$\begin{array}{lll}
 \hat{x}_{t+1|t} & := & \mathbb{E}(x_{t+1}|y_0, \dots, y_t) \\
 P_{t+1|t} & = & \mathbb{E}(\tilde{x}_{t+1|t}\tilde{x}_{t+1|t}^T|y_0, \dots, y_t) \\
 \tilde{x}_{t+1|t} & = & x_{t+1} - \hat{x}_{t+1|t}
 \end{array}
 \Rightarrow
 \begin{array}{lll}
 \hat{x}_{t+1|t+1} & := & \mathbb{E}(x_{t+1}|y_0, \dots, y_{t+1}) \\
 P_{t+1|t+1} & = & \mathbb{E}(\tilde{x}_{t+1|t+1}\tilde{x}_{t+1|t+1}^T|y_0, \dots, y_{t+1}) \\
 \tilde{x}_{t+1|t+1} & = & x_{t+1} - \hat{x}_{t+1|t+1}
 \end{array}$$

Distribution of y_{t+1} given $y_0, \dots, y_t$:

$$\begin{aligned}
 \hat{y}_{t+1|t} &:= \mathbb{E}(y_{t+1}|y_0, \dots, y_t) \\
 &= \mathbb{E}(Cx_{t+1} + v_{t+1}|y_0, \dots, y_t) = C\hat{x}_{t+1|t} \\
 \mathbb{E} \left[(y_{t+1} - \hat{y}_{t+1|t})(y_{t+1} - \hat{y}_{t+1|t})^T | y_0, \dots, y_t \right] \\
 &= \mathbb{E} \left[(Cx_{t+1} + v_{t+1} - C\hat{x}_{t+1|t})(Cx_{t+1} + v_{t+1} - C\hat{x}_{t+1|t})^T | y_0, \dots, y_t \right] \\
 &= \mathbb{E} \left[(C(x_{t+1} - \hat{x}_{t+1|t}) + v_{t+1})(C(x_{t+1} - \hat{x}_{t+1|t}) + v_{t+1})^T | y_0, \dots, y_t \right] \\
 &= CP_{t+1|t}C^T + R \\
 \mathbb{E} \left[(y_{t+1} - \hat{y}_{t+1|t})(x_{t+1} - \hat{x}_{t+1|t})^T | y_0, \dots, y_t \right] \\
 &= \mathbb{E} \left[(Cx_{t+1} + v_{t+1} - C\hat{x}_{t+1|t})(x_{t+1} - \hat{x}_{t+1|t})^T | y_0, \dots, y_t \right] \\
 &= CP_{t+1|t}
 \end{aligned}$$

Measurement Update (2)

x_{t+1} and y_{t+1} have the conditional joint multivariate Gaussian distribution

$$P(x_{t+1}, y_{t+1} | y_0, \dots, y_t) \sim \mathcal{N}(\mu, \Sigma),$$

where

$$\mu = \begin{pmatrix} \hat{x}_{t+1|t} \\ C\hat{x}_{t+1|t} \end{pmatrix}, \quad \Sigma = \begin{pmatrix} P_{t+1|t} & P_{t+1|t}C^T \\ CP_{t+1|t} & CP_{t+1|t}C^T + R \end{pmatrix}$$

- $P(x_{t+1}, y_{t+1} | y_0, \dots, y_t) \sim \mathcal{N}(\mu, \Sigma)$, where

$$\mu = \begin{pmatrix} \mu_x \\ \mu_y \end{pmatrix}, \quad \Sigma = \begin{pmatrix} \Sigma_{xx} & \Sigma_{xy} \\ \Sigma_{yx} & \Sigma_{yy} \end{pmatrix}$$

- $P(x_{t+1} | y_0, \dots, y_t, y_{t+1}) \sim \mathcal{N}(\mu_{x|y}, \Sigma_{x|y})$, where

$$\begin{aligned} \mu_{x|y} &= \mu_x + \Sigma_{xy}\Sigma_{yy}^{-1}(y - \mu_y) \\ \Sigma_{x|y} &= \Sigma_{xx} - \Sigma_{xy}\Sigma_{yy}^{-1}\Sigma_{yx} \end{aligned}$$

$P(x_{t+1} | y_0, \dots, y_t, y_{t+1}) \sim \mathcal{N}(\hat{x}_{t+1|t+1}, P_{t+1|t+1})$, where

$$\begin{aligned} \hat{x}_{t+1|t+1} &= \hat{x}_{t+1|t} + P_{t+1|t}C^T(CP_{t+1|t}C^T + R)^{-1}(y_{t+1} - C\hat{x}_{t+1|t}) \\ P_{t+1|t+1} &= P_{t+1|t} - P_{t+1|t}C^T(CP_{t+1|t}C^T + R)^{-1}CP_{t+1|t} \end{aligned}$$

Kalman Filter

- Dynamic update

$$\begin{aligned}\hat{x}_{t+1|t} &= A\hat{x}_{t|t} \\ P_{t+1|t} &= AP_{t|t}A^T + GQG^T\end{aligned}$$

- Measurement update

$$\begin{aligned}\hat{x}_{t+1|t+1} &= \hat{x}_{t+1|t} + K_{t+1}(y_{t+1} - C\hat{x}_{t+1|t}) \\ P_{t+1|t+1} &= P_{t+1|t} - K_{t+1}CP_{t+1|t}\end{aligned}$$

- Kalman gain

$$K_{t+1} = P_{t+1|t}C^T(CP_{t+1|t}C^T + R)^{-1}$$

- Dynamic model: $x_{t+1} = Ax_t + Gw_t$
- Measurement model: $y_t = Cx_t + v_t$
- Initial state distribution: $x_0 \sim \mathcal{N}(\mu_0, \Sigma_0)$
- Noises: $w_t \sim \mathcal{N}(0, Q)$, $v_t \sim \mathcal{N}(0, R)$. w_t , v_t , and x_t are independent.

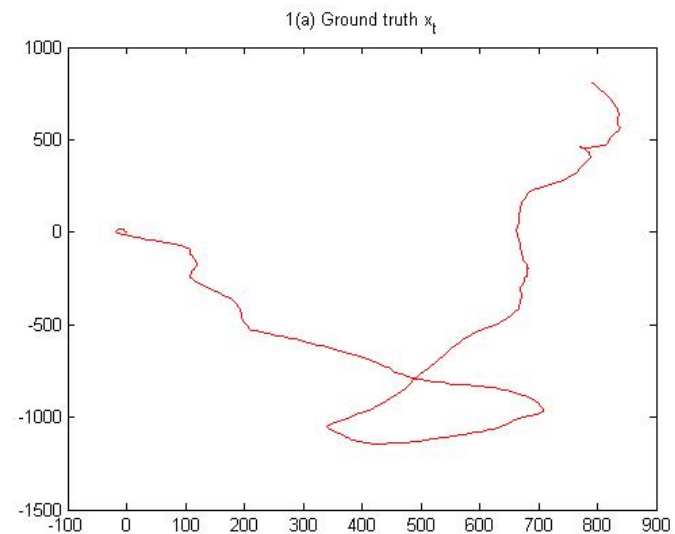
Example (1)

A particle moving in a plane under random forces and damping.

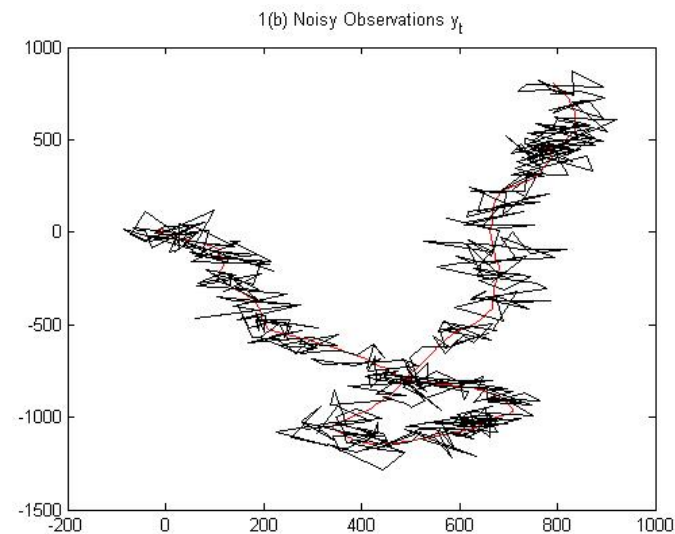
- State of the particle: $x = [x^1, \dot{x}^1, x^2, \dot{x}^2]^T$
- (x^1, x^2) : position of the particle; $(\dot{x}^1, \dot{x}^2)$: velocity of the particle
- $x_{t+1}^i = x_t^i + \dot{x}_t^i$
- $\dot{x}_{t+1}^i = 0.98\dot{x}_t^i + w_t^i$
- Dynamic model: $x_{t+1} = Ax_t + Gw_t$
- Measurement model: $y_t = Cx_t + v_t$

$$A = \begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 0.98 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0.98 \end{bmatrix} \quad G = \begin{bmatrix} 0 & 0 \\ 1 & 0 \\ 0 & 0 \\ 0 & 1 \end{bmatrix} \quad C = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

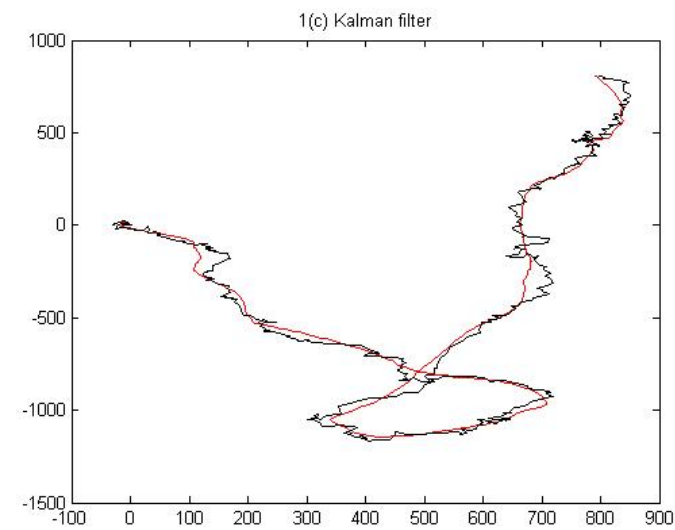
Example (2)



Ground Truth



Noisy
Measurements



Kalman filter

Extended Kalman Filters

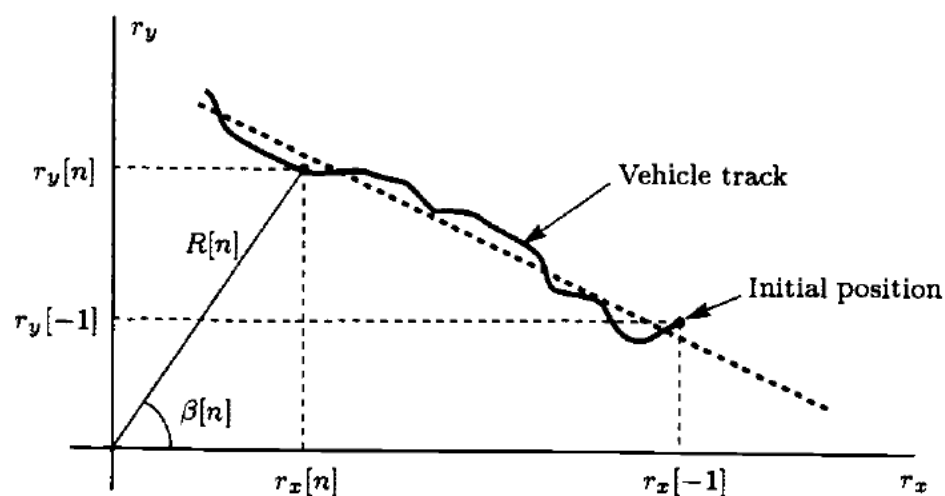
Particle Filters

OTHER FILTERING ALGORITHMS

Extended Kalman Filters (1)

- When the dynamic or measurement is nonlinear, previous results cannot be used.
- But we can still apply the previous results by linearizing the nonlinear functions.

Vehicle tracking example



Vehicle's position: (r_x, r_y)

Nonlinear measurement model

$$\hat{R}[n] = R[n] + w_R[n]$$

$$\hat{\beta}[n] = \beta[n] + w_\beta[n]$$

$$R[n] = \sqrt{r_x^2[n] + r_y^2[n]} \quad \text{Range}$$

$$\beta[n] = \arctan \frac{r_y[n]}{r_x[n]} \quad \text{Bearing (angle)}$$

Source: Steven M. Kay, "Fundamentals of Statistical Signal Processing, Volume I: Estimation Theory", Prentice Hall, 1993.

Extended Kalman Filters (2)

Original nonlinear system:

$$\begin{aligned} \mathbf{s}[n] &= \mathbf{a}(\mathbf{s}[n-1]) + \mathbf{B}\mathbf{u}[n] & \mathbf{a}(\cdot) \text{ and } \mathbf{h}(\cdot) \text{ are nonlinear vector valued functions} \\ \mathbf{x}[n] &= \mathbf{h}(\mathbf{s}[n]) + \mathbf{w}[n] & \mathbf{u}[n] \sim \mathcal{N}(\mathbf{0}, \mathbf{Q}) \quad \mathbf{w}[n] \sim \mathcal{N}(\mathbf{0}, \mathbf{C}[n]) \end{aligned}$$

First-order Taylor expansion:

$$\begin{aligned} \mathbf{a}(\mathbf{s}[n-1]) &\approx \mathbf{a}(\hat{\mathbf{s}}[n-1|n-1]) \\ &\quad + \left. \frac{\partial \mathbf{a}}{\partial \mathbf{s}[n-1]} \right|_{\mathbf{s}[n-1]=\hat{\mathbf{s}}[n-1|n-1]} (\mathbf{s}[n-1] - \hat{\mathbf{s}}[n-1|n-1]) \\ \mathbf{h}(\mathbf{s}[n]) &\approx \mathbf{h}(\hat{\mathbf{s}}[n|n-1]) + \left. \frac{\partial \mathbf{h}}{\partial \mathbf{s}[n]} \right|_{\mathbf{s}[n]=\hat{\mathbf{s}}[n|n-1]} (\mathbf{s}[n] - \hat{\mathbf{s}}[n|n-1]) . \end{aligned}$$

Let

$$\begin{aligned} \mathbf{A}[n-1] &= \left. \frac{\partial \mathbf{a}}{\partial \mathbf{s}[n-1]} \right|_{\mathbf{s}[n-1]=\hat{\mathbf{s}}[n-1|n-1]} \\ \mathbf{H}[n] &= \left. \frac{\partial \mathbf{h}}{\partial \mathbf{s}[n]} \right|_{\mathbf{s}[n]=\hat{\mathbf{s}}[n|n-1]} \end{aligned}$$

Linearized dynamic and measurement models

$$\begin{aligned} \mathbf{s}[n] &= \mathbf{A}[n-1]\mathbf{s}[n-1] + \mathbf{B}\mathbf{u}[n] + (\mathbf{a}(\hat{\mathbf{s}}[n-1|n-1]) - \mathbf{A}[n-1]\hat{\mathbf{s}}[n-1|n-1]) \\ \mathbf{x}[n] &= \mathbf{H}[n]\mathbf{s}[n] + \mathbf{w}[n] + (\mathbf{h}(\hat{\mathbf{s}}[n|n-1]) - \mathbf{H}[n]\hat{\mathbf{s}}[n|n-1]) . \end{aligned}$$

Extended Kalman Filters (3)

Prediction:

$$\hat{\mathbf{s}}[n|n-1] = \mathbf{a}(\hat{\mathbf{s}}[n-1|n-1]). \quad (13.67)$$

Minimum Prediction MSE Matrix ($p \times p$):

$$\mathbf{M}[n|n-1] = \mathbf{A}[n-1]\mathbf{M}[n-1|n-1]\mathbf{A}^T[n-1] + \mathbf{BQB}^T. \quad (13.68)$$

Kalman Gain Matrix ($p \times M$):

$$\mathbf{K}[n] = \mathbf{M}[n|n-1]\mathbf{H}^T[n] (\mathbf{C}[n] + \mathbf{H}[n]\mathbf{M}[n|n-1]\mathbf{H}^T[n])^{-1}. \quad (13.69)$$

Correction:

$$\hat{\mathbf{s}}[n|n] = \hat{\mathbf{s}}[n|n-1] + \mathbf{K}[n](\mathbf{x}[n] - \mathbf{h}(\hat{\mathbf{s}}[n|n-1])). \quad (13.70)$$

Minimum MSE Matrix ($p \times p$):

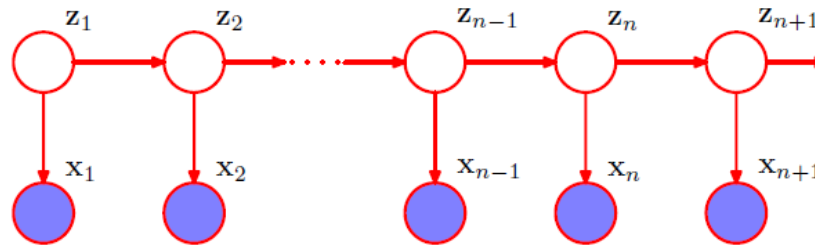
$$\mathbf{M}[n|n] = (\mathbf{I} - \mathbf{K}[n]\mathbf{H}[n])\mathbf{M}[n|n-1] \quad (13.71)$$

where

$$\begin{aligned} \mathbf{A}[n-1] &= \left. \frac{\partial \mathbf{a}}{\partial \mathbf{s}[n-1]} \right|_{\mathbf{s}[n-1] = \hat{\mathbf{s}}[n-1|n-1]} \\ \mathbf{H}[n] &= \left. \frac{\partial \mathbf{h}}{\partial \mathbf{s}[n]} \right|_{\mathbf{s}[n] = \hat{\mathbf{s}}[n|n-1]}. \end{aligned}$$

Particle Filters (1)

- For dynamical systems which is not linear-Gaussian, a particle filter uses a sampling method to approximate the posterior distribution.
- Particle filtering is also known as *sequential Monte Carlo* or a *condensation* algorithm in computer vision.



$$\begin{aligned}\mathbb{E}[f(\mathbf{z}_n)] &= \int f(\mathbf{z}_n) p(\mathbf{z}_n | \mathbf{X}_n) d\mathbf{z}_n \\ &= \int f(\mathbf{z}_n) p(\mathbf{z}_n | \mathbf{x}_n, \mathbf{X}_{n-1}) d\mathbf{z}_n \\ &= \frac{\int f(\mathbf{z}_n) p(\mathbf{x}_n | \mathbf{z}_n) p(\mathbf{z}_n | \mathbf{X}_{n-1}) d\mathbf{z}_n}{\int p(\mathbf{x}_n | \mathbf{z}_n) p(\mathbf{z}_n | \mathbf{X}_{n-1}) d\mathbf{z}_n} \\ &\simeq \sum_{l=1}^L w_n^{(l)} f(\mathbf{z}_n^{(l)})\end{aligned}$$

- $\mathbf{z}_n$: state at time n
- $\mathbf{X}_n = (x_1, \dots, x_n)$: measurements
- $\mathbf{z}_n^{(l)}$: a sample from $p(\mathbf{z}_n | \mathbf{X}_{n-1})$
- $w_n^{(l)}$: sampling weight

$$w_n^{(l)} = \frac{p(\mathbf{x}_n | \mathbf{z}_n^{(l)})}{\sum_{m=1}^L p(\mathbf{x}_n | \mathbf{z}_n^{(m)})}$$

Source: Christopher M. Bishop, "Pattern Recognition and Machine Learning", Springer, 2007.

Particle Filters (2)

